

- Debugging
- Iteration
- Hallucination Control
- Task: 20 weak prompts ko improve karna

Main har example mein **Weak Prompt** → **Problems** → **Improved Prompt** → **Why it is better** dunga, taa-ke assignment sirf prompts ki list na ho balki academically strong bhi lage.

20 Weak Prompts → Improved Prompts

#	Weak Prompt	Improved Prompt
1	Explain AI.	Act as an AI instructor. Explain Artificial Intelligence to a beginner in simple language. Cover: definition, major subfields, real-world applications, and limitations. Use 2-3 concrete examples. Avoid unnecessary jargon.
2	Tell me about machine learning.	Explain Machine Learning for a university beginner. Define ML, explain supervised, unsupervised, and reinforcement learning, and provide one practical example for each. Clearly distinguish ML from traditional rule-based programming.
3	Write Python code for login.	Act as a Python developer. Create a simple educational login system in Python using functions. Validate username and password input, handle invalid credentials safely, and explain the code. Do not store passwords in plaintext in a production recommendation.
4	Why is my code not working?	Debug the following Python code. First identify the exact error(s), then explain the root cause, provide a corrected version, and briefly explain each modification. Do not assume information that is not present in the code. Code: [PASTE CODE]
5	Fix this code.	Analyze the code below for syntax, logical, and runtime errors. For each issue, provide: (1) location, (2) problem, (3) reason, and (4) correction. Then provide the complete corrected code. Preserve the original intended behavior unless a change is necessary.

#	Weak Prompt	Improved Prompt
6	Make my prompt better.	Improve the following prompt for clarity, specificity, reliability, and output consistency. Preserve its original objective. Identify ambiguous instructions first, then provide the improved prompt and briefly explain the major changes. Prompt: [PROMPT]
7	Give me facts about AI.	Provide 10 factual statements about Artificial Intelligence. For each statement, distinguish established facts from estimates or interpretations. Do not invent statistics, studies, dates, or citations. If a claim cannot be verified from your available knowledge, explicitly mark it as uncertain.
8	Who is the best AI model?	Compare major LLM categories based on reasoning, coding, context handling, speed, cost, and tool use. Do not declare a universal "best" model because performance depends on the task. Explain which characteristics matter for different use cases.
9	Explain transformers.	Explain the Transformer architecture in five stages: tokenization/input representation, positional information, self-attention, feed-forward processing, and output generation. Use a small conceptual example and explain why Transformers became important for modern LLMs.
10	Summarize this.	Summarize the following text in 5 bullet points. Preserve the author's main claims, important evidence, limitations, and conclusions. Do not introduce information that is absent from the source. If an important statement is ambiguous, retain the uncertainty.
11	Research RAG and tell me everything.	Research Retrieval-Augmented Generation (RAG). Structure the answer into: definition, architecture, retrieval pipeline, embeddings, vector databases, reranking, generation, evaluation, common failure modes, and current limitations. Clearly distinguish established concepts from emerging practices. Do not fabricate papers, benchmarks, or citations.

#	Weak Prompt	Improved Prompt
12	Create a business plan for an AI SaaS.	Act as an AI SaaS product strategist. Develop a business-plan outline for an AI SaaS product. Include target customer, problem, solution, value proposition, core features, architecture, acquisition strategy, pricing assumptions, risks, competitors, and MVP scope. Clearly label assumptions instead of presenting them as facts.
13	Tell me why my AI gives wrong answers.	Diagnose possible causes of incorrect LLM responses. Categorize causes into prompting, context quality, retrieval, model limitations, tool errors, data quality, and evaluation problems. For each category, provide a debugging test and a practical mitigation.
14	Make AI stop hallucinating.	Design a hallucination-control strategy for an LLM application. Require the model to distinguish known information, retrieved evidence, assumptions, and uncertainty. Instruct it to avoid unsupported claims and explicitly state when evidence is insufficient. Include a verification and citation policy.
15	Answer this question accurately.	Answer the question using only information you can support. Before giving factual claims, check whether the necessary information is available in the supplied context. If evidence is insufficient, say "Insufficient information" rather than guessing. Separate facts from assumptions.
16	Improve this answer again.	Iteratively improve the answer below. In each iteration, identify one concrete weakness, revise it, and verify that the revision does not introduce unsupported claims. Stop when the answer is clear, complete, internally consistent, and aligned with the original requirements. Return only the final version plus a brief list of changes.
17	Find the mistake in my reasoning.	Critically inspect the reasoning below. Identify unsupported assumptions, logical gaps, contradictions, factual errors, and missing evidence. Do not assume the conclusion is wrong; determine whether each step follows from the previous one. Provide corrected reasoning where necessary.

#	Weak Prompt	Improved Prompt
18	Create a prompt for an AI agent.	Design a production-oriented prompt for an AI agent that can use tools. Define: role, objective, available tools, tool-selection rules, input requirements, execution constraints, error handling, hallucination controls, output schema, and stopping conditions. The agent must never claim a tool action succeeded unless the tool result confirms it.
19	Analyze this data and give insights.	Analyze the provided dataset. First describe the dataset's structure and identify missing values, duplicates, anomalies, and possible data-quality problems. Then calculate relevant descriptive statistics and identify meaningful patterns. Separate observed results from interpretations and avoid causal claims unless supported by the data.
20	Give me the correct answer and don't make anything up.	Provide the most reliable answer possible. Use the supplied evidence as the primary source. For every important factual claim, determine whether it is directly supported, reasonably inferred, or uncertain. Never fabricate facts, sources, citations, numbers, or events. If the evidence is insufficient, explicitly state what information is missing. Before finalizing, perform a consistency check against the original question.

What changed?

In weak prompts, common problems were:

1. **Ambiguity** — model ko exactly pata nahi hota ke output kaisa chahiye.
2. **No role definition** — task ka context missing hota hai.
3. **No constraints** — model unnecessary information generate kar sakta hai.
4. **No verification** — factual errors easily pass ho jate hain.
5. **No uncertainty handling** — model guessing kar sakta hai.
6. **No output structure** — har run par different answer aa sakta hai.
7. **No debugging protocol** — sirf "fix it" kehne se root cause identify nahi hota.
8. **No iteration criteria** — improvement ka koi measurable stopping condition nahi.
9. **Hallucination vulnerability** — unsupported facts confidently generate ho sakte hain.
10. **Facts aur assumptions mix hona** — especially research/business prompts mein dangerous hai.

Core lesson

A strong prompt sirf "AI ko kya karna hai" nahi batata. Woh ye bhi define karta hai:

Task → Context → Constraints → Process → Evidence → Uncertainty → Verification → Output Format

Yahi pattern tumhari assignment mein **Debugging + Iteration + Hallucination Control** ko properly demonstrate karega.